

Data Mining & Knowledge Engineering

Sukjaem/Kim/Khan

William Bank Sukjaem (18029208), Juchang Kim (22180242), Raihan Zaid Khan (23202181)

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I. CONTRIBUTIONS

A. William

William conducted the analysis and coding for the Depression Reddit dataset, covering both unprocessed and preprocessed versions in Google Colab. He implemented and compared Naïve Bayes, Regularised Neural Network, and CNN models, optimising hyperparameters to improve classification performance. William also initiated the project documentation, drafting the abstract, introduction, methodology, and data description, and contributed to the results and discussion sections for the Depression dataset.

B. Juchang

Juchang led the documentation and refinement of the Covid News Articles notebooks, including both raw and preprocessed datasets. He revised the methodology and experimental workflow, produced and interpreted results, and authored key sections such as discussion, conclusion, and reflection. Juchang also conducted literature research to explain model behaviours and integrated references related to Naïve Bayes, SVM, and Neural Network classifiers.

C. Raihan

Raihan initially explored additional algorithms on other datasets to assess their suitability. Raihan developed and tested the Covid News Articles notebooks on Google Colab for both unprocessed and preprocessed datasets. He implemented Naïve Bayes, Linear SVM, and MLP models, ensuring consistent preprocessing, TF-IDF feature extraction, and performance reporting. Raihan prepared summary tables and visualisations, streamlined the notebook structure for reproducibility, and contributed to refining the documentation and discussion of model performance and efficiency.

Abstract

This paper aims to compare the performance and accuracy of text classification algorithms to two distinct datasets: *Depression Reddit Dataset* and the *COVID New Article Dataset*. For both datasets, Naïve Bayes and Neural Network classifier algorithms were used as baseline models. In addition to those algorithms, we applied Convolutional Neural Network (CNN) to the *Depression Reddit Dataset*, while Support Vector Machine (SVM) was applied to the *COVID Dataset*. Model evaluation was done using standard metrics such as accuracy, precision, recall and f1-score to measure and compare the performance of each algorithm.

II. INTRODUCTION

Text classification involves automatically categorising text, while it shares the same goal as classifying numerical data, it is more complex due to text must first be converted into a numerical format. The core differences is in data representation. In the context of text classification, text must first be converted into a numerical format such as TF-IDF or word embeddings, which inherently introduces high dimensionality and sparsity. The experiments is to compare the performance of various text classification algorithms on two distinct datasets. The algorithms include Naïve Bayes, a fast probabilistic model; Multi-Layer Perceptron (MLP), a neural network that learns complex non-linear boundaries; Convolutional Neural Network (CNN), which specialises in detecting informative local phrase patterns; and Support Vector Machine (SVM), an algorithm that finds the optimal hyperplane to separate classes. The similarity is distinctly in their shared classification goal, but they differ in their approaches. Naïve Bayes relies on probability and feature independence, MLP uses interconnected layers of neurons, SVM focuses on maximising the margin between classes, and CNN uniquely captures local word order, giving it an advantage in understanding contextual phrases.

III. METHODOLOGY

The methodology involved an initial phase of experimentation and exploration across several available datasets, including the COVID News Articles, Depression Reddit, Fake and Real News, and Emotional Datasets. From these, two datasets—COVID News Articles and Depression Reddit which were selected based on their consistency, interpretability, and suitability for comparative text classification analysis. The COVID dataset represents a multi-class sentiment corpus containing five (later merged into three) sentiment categories, while the Depression Reddit dataset is a binary classification task distinguishing between depressed and non-depressed user comments. Both datasets are relatively concise, text-focused, and well-suited for evaluating sentiment and emotion recognition algorithms.

Multiple algorithms were evaluated during the exploration stage, including Naïve Bayes, Feedforward Neural Network (Multilayer Perceptron), Support Vector Machine (SVM), and Convolutional Neural Network (CNN). Among these, Naïve Bayes and the Feedforward Neural Network were used as baseline models. These algorithms were chosen to represent both traditional probabilistic and modern neural approaches, Naïve Bayes for its efficiency and interpretability, SVM for its strength in handling high-dimensional sparse data, MLP for learning non-linear relationships, and CNN for its ability to capture local word patterns and contextual cues in text. Each algorithm was applied to both unprocessed (raw) and preprocessed text data to analyse the effect of cleaning and normalisation on classification performance.

Preprocessing steps included lowercasing, removing URLs and punctuation, expanding contractions, filtering non-alphabetic tokens, and applying lemmatisation to normalise word forms. These operations were implemented to reduce textual noise, unify structure, and ensure that semantically meaningful words, particularly sentiment-related terms—were preserved.

Following the initial experiments, fine-tuning and optimisation were conducted to improve model performance and establish consistent benchmarks for comparison. For the preprocessed datasets, hyperparameters such as learning rate, regularisation strength, number of hidden units, and dropout rate were systematically adjusted to maximise accuracy while preventing overfitting. All models were trained using an 80/20 stratified train–test split, with a small subset of the training data reserved for validation. This methodological design ensured a consistent, unbiased, and reproducible framework for evaluating the performance of different algorithms on text-based sentiment and emotion classification tasks.

IV. DATA DESCRIPTION

The chosen datasets that is used for thorough analysis are the *Depression Reddit Dataset* and the *COVID New Articles Dataset*, which involves a relatively small set of dimensions, of a maximum of 6 columns, that is mostly cleaned. These datasets are user collected data from Kaggle where *Depression Reddit Dataset* is raw web-scraped text from various subreddits and *COVID New Articles Dataset* is collected from half a million articles from the internet over two years of the COVID-19 pandemic starting from 2020 (Kaggle, 2022). Metadata of the datasets includes some classifier that identifies a text string as a representation of sorts. In the case of the chosen datasets, *Depression Reddit Dataset* includes two columns *clean_text*, which is extracted text string of user comments; and *is_depressed*, which is a numeric binary classifier which determines whether the associated comment has been identified as depressed or not. As for the *COVID New Articles Dataset*, it consists of six

columns being *UserName* and *ScreenName*, which are user ID numbers that represent author of the post; *Location*, which has a string description of the city location of the post; *TweetAt*, which has the date of the tweet; *OriginalTweet*, displaying the string text of the post; and *Sentiment*, which displays the sentimental categorical classifier of the post that consists of five value types: Extremely Negative, Negative, Neutral, Positive and Extremely Positive.

V. EXPERIMENTS

Four datasets were initially reviewed: COVID News Articles, Depression Reddit, Fake and Real News, and Emotional Datasets. Among these, the COVID News Articles and Depression Reddit datasets were selected for detailed experimentation due to their clear sentiment labels, balanced structure, and suitability for evaluating text classification algorithms. Both datasets represent concise, user-generated language which are Tweets and Reddit comments—making them ideal for sentiment and emotion detection tasks.

In each dataset, only the relevant text and label columns were retained (e.g., *OriginalTweet* and *Sentiment* for COVID data, *comments* and *is_depression* for Reddit). The data were stratified using an 80/20 train–test split to preserve class proportions and ensure fair model evaluation. To maintain consistency, both datasets were processed through the same experimental pipeline described in Section II.

Three algorithms, Naïve Bayes (NB), Support Vector Machine (SVM), and a Feedforward Neural Network (Multilayer Perceptron, MLP), were applied to the COVID dataset, while NB, MLP, and a Convolutional Neural Network (CNN) were used for the Depression Reddit dataset. This selection ensured a balanced comparison between traditional machine learning and deep learning approaches.

For NB and SVM, TF-IDF (Term Frequency–Inverse Document Frequency) vectorisation was employed to convert text into sparse feature matrices, as these models perform effectively on high-dimensional linear representations. For neural models (MLP and CNN), tokenisation and embedding layers were used to create dense, trainable numerical representations that capture contextual word relationships. The CNN was specifically included for the Reddit dataset to model local semantic dependencies such as emotional phrases (e.g., “feel hopeless” or “can’t sleep”).

All models were tuned to achieve optimal performance based on cross-validation results. For the COVID dataset, Grid-SearchCV was applied to optimise parameters such as the smoothing factor (α) for NB and the regularisation constant (C) for SVM. Neural networks were tuned for learning rate, hidden layer size, dropout, and batch size. Preprocessing and text normalisation steps (Section II) were performed before rerunning the same models on the cleaned datasets, allowing direct comparison between raw and preprocessed performance outcomes. This experimental design enabled a structured evaluation of how algorithm choice, feature representation, and text preprocessing collectively influence performance across two distinct text-based sentiment classification tasks.

A. Dataset, Algorithm Justification and Assumption

Four datasets were initially reviewed for experimentation: COVID News Articles, Depression Reddit, Fake and Real News, and Emotional Datasets. Among these, the COVID News Articles and Depression Reddit datasets were selected due to their manageable size, linguistic consistency, and relevance to sentiment and emotion classification. Both focus on predicting emotional tone or polarity from short text, making them ideal for evaluating the performance of different machine learning and deep learning models. Compared to large-scale news corpora, these datasets are more computationally efficient to process while still offering linguistic diversity and emotionally rich language.

A key consideration in dataset selection was text length and domain uniformity. Tweets and Reddit comments typically contain concise, emotionally expressive language, providing a realistic yet controlled setting for testing how algorithms capture sentiment cues from short, unstructured text. To ensure fair and unbiased evaluation, both datasets were split using stratified sampling to maintain balanced class distributions across training and testing subsets.

Three main classification algorithms were applied to the COVID dataset which are Naïve Bayes (NB), Support Vector Machine (SVM), and a Feedforward Neural Network (Multilayer Perceptron, MLP). While the Depression Reddit dataset utilised NB, MLP, and a Convolutional Neural Network (CNN). This selection provided a comprehensive comparison between traditional machine learning and modern neural network approaches.

Naïve Bayes (NB): NB is a probabilistic classifier based on Bayes’ Theorem that assumes feature independence. Although this assumption rarely holds in natural language, NB performs effectively on high-dimensional and sparse textual data. It provides a strong baseline for sentiment classification, efficiently leveraging word frequency distributions while maintaining

interpretability and low computational cost (Manning & Schütze, 1999).

Support Vector Machine (SVM): SVM was applied to the COVID dataset as a linear classifier that identifies the optimal hyperplane separating sentiment classes. Its strength lies in handling high-dimensional TF-IDF features and its robustness to overfitting through regularisation. SVM has consistently achieved high performance in text classification tasks (Joachims, 1998). The model assumes that data can be linearly separated in the transformed feature space, though kernel methods can be introduced to address non-linear relationships.

Neural Network (MLP): A feedforward Neural Network (Multilayer Perceptron) was implemented for both datasets to capture complex, non-linear relationships between textual features and sentiment classes. Using TF-IDF representations as input, the MLP model learns hierarchical patterns that go beyond linear decision boundaries. Regularisation techniques such as dropout and early stopping were applied to prevent overfitting and ensure stable generalisation across datasets.

Convolutional Neural Network (CNN): The CNN model was employed for the Depression Reddit dataset due to its strong ability to detect local linguistic and contextual patterns. Through the use of convolutional filters, CNNs capture dependencies in word sequences—identifying emotional phrases such as “feel hopeless” or “can’t sleep.” This architecture assumes that neighbouring words contribute more strongly to meaning than distant ones, aligning with how emotional expressions manifest in human language (Kim, 2014).

Before model training, each dataset was transformed into numerical representations suitable for its corresponding algorithm. NB, SVM, and MLP employed TF-IDF vectorisation to generate weighted feature matrices, while the CNN used tokenisation and word embedding layers to convert text sequences into dense, trainable representations that preserve contextual relationships. This unified experimental design enabled direct and consistent comparison between classical statistical models and deep learning approaches in sentiment and emotion classification.

B. Covid News Articles Dataset

a) *Raw 5-Class Split.*: The COVID News Articles dataset originally provided separate training and testing files. To ensure a balanced and unbiased data distribution, both files were merged and re-split using stratified sampling. This approach preserved the proportion of each sentiment category across the new partitions, maintaining consistency between training and testing data. The dataset includes five sentiment labels—*Extremely Positive*, *Positive*, *Neutral*, *Negative*, and *Extremely Negative*—representing a broad emotional spectrum in public reactions to COVID-related news. To prepare the data for model training, textual inputs were transformed into numerical representations using TF-IDF (Term Frequency–Inverse Document Frequency), which assigns higher weights to informative words and down-weights common terms. This vectorisation approach was particularly suitable for traditional machine learning models such as Naïve Bayes (NB) and Support Vector Machine (SVM), while also serving as input for the baseline Neural Network (MLP) to ensure consistent comparison. Hyperparameter tuning for NB and SVM was performed using `GridSearchCV`, optimising the maximum number of features, n-gram range, and smoothing or regularisation parameters (α for NB and C for SVM). The optimisation aimed to identify the most effective configuration for achieving high accuracy without overfitting. For the MLP, TF-IDF vectors were also used as input features, with label encoding applied to convert categorical sentiment strings into integer targets compatible with neural training. Early stopping and dropout were incorporated to mitigate overfitting and improve generalisation.

b) *Preprocessed 3-Class Split.*: After completing the experiments with the raw dataset, a simplified and more balanced 3-class version was created. The *Extremely Positive* and *Positive* labels were merged into a single *Positive* category, while *Extremely Negative* and *Negative* were combined into *Negative*; *Neutral* remained unchanged. This reduction addressed class imbalance and improved the generalisability of the models. Comprehensive text preprocessing and normalisation were applied to reduce noise and improve linguistic consistency. These steps included lowercasing, expanding contractions (e.g., “can’t” → “cannot”), replacing emojis and URLs with descriptive placeholders, removing punctuation and special characters, splitting hashtags into component words, tokenising text, removing very short tokens, and applying lemmatisation to standardise word forms. This process improved semantic coherence and reduced redundancy within the corpus. The Naïve Bayes, SVM, and MLP models were retrained using the preprocessed dataset under identical hyperparameter settings to ensure fair comparison with the raw text results. This stage enabled analysis of how linguistic normalisation and label simplification influence classification performance. The comparison between the raw and preprocessed results revealed that data cleaning and class merging led to more stable, interpretable, and generalisable model outcomes.

c) *Naïve Bayes (MultinomialNB).*: The Multinomial Naïve Bayes classifier was used as a baseline model for both the raw and preprocessed datasets. Despite its independence assumption between features, NB is computationally efficient and

effective in high-dimensional text data. The model’s simplicity and interpretability make it ideal for benchmarking more complex classifiers. Cross-validation ensured reliable parameter selection and avoided feature leakage, providing a transparent and replicable baseline for performance comparison.

d) Support Vector Machine (LinearSVC).: The Linear SVM model served as a complementary baseline to NB. By learning a hyperplane that maximises the margin between sentiment classes, SVM is particularly effective in high-dimensional spaces such as TF-IDF feature matrices. Regularisation through the C parameter controlled overfitting, while cross-validation identified the best decision boundary. Although some sentiment overlap (e.g., between *Negative* and *Extremely Negative*) reduces perfect separability, the linear kernel remained competitive and robust across both raw and preprocessed data.

e) Neural Network (MLP).: A feedforward Neural Network (Multilayer Perceptron) was implemented to explore non-linear decision boundaries beyond the scope of NB and SVM. The architecture consisted of successive dense layers with ReLU activation, dropout layers (0.5 rate), and a softmax output layer. Hyperparameters such as layer size (512 and 256 units), learning rate (10^{-3}), and batch size (128) were tuned through experimentation. The model’s ability to capture subtle sentiment differences (e.g., between “positive” and “neutral”) demonstrated the benefit of non-linear feature combinations, though it required greater computational effort and careful regularisation.

C. Depression Reddit Raw Dataset Preparation

a) Unprocessed Dataset: The Depression Reddit Dataset contained user-submitted posts labelled under a binary variable `is_depression`, indicating whether a comment expresses depressive or non-depressive language. Like the *Covid New Articles* dataset, an 80/20 stratified split was done preserving the ratio of both classes across the partitions and TF-IDF vectorisation was applied.

For the Neural Network models, the text was processed through keras tokenisation and sequence padding, where each word was converted into a numerical index and padded to a consistent sequence length. The tokenised sequences were then passed through embedding layers to enable the model to learn distributed word representations. Since the labels were already binary, additional encoding was unnecessary. To ensure that performance differences were driven by data and model behaviour rather than training irregularities, consistent training safeguards by using early stopping.

Several data integrity checks were performed. These checks included detecting duplicate and overlapping samples between the training and test sets, as well as identifying duplicate comments within the training set. The distribution of comment lengths was also examined, including counting extremely short comments (3 words or fewer). While these checks provided valuable insights into the dataset’s composition, this is addressed in the preprocessing of the dataset.

b) Preprocessed Dataset: Based on the initial unprocessed *Depression Reddit Dataset*, preprocessing procedures are done before any feature extraction or model training or comprehensive data-integrity validation was conducted to ensure the reliability of the dataset and to prevent artificial inflation of model performance. This included identifying and removing duplicate entries within the training data and detecting any overlap between the training and testing partitions. In particular, comments appearing in both splits were eliminated to avoid data leakage, thereby ensuring that evaluation metrics reflected genuine generalisation on unseen samples.

Following these integrity checks, a preprocessing pipeline was applied to refine linguistic quality while preserving the emotional and psychological content of the text. The preprocessing procedure aimed to improve consistency and reduce textual noise without diminishing the expressive signals associated with depressive sentiment. The core of this pipeline was a custom rule-based cleaning function, which performed several deterministic operations such as converting all text to lowercase, removing URLs, filtering non-alphabetic symbols and numeric characters, and normalising whitespace.

A key component of this pipeline was its selective stopwords filtering strategy. While common English stopwords were removed to reduce lexical redundancy, a predefined subset of emotionally charged words—such as *feel*, *hopeless*, *tired*, and *empty*—was deliberately retained. This design choice preserved affective tokens that often serve as salient indicators of psychological distress, ensuring that the dataset remained both clean and semantically faithful to its depressive context.

After these preprocessing and cleaning steps, the data was maintained in its stratified 80/20 train–test split, using identical randomisation and stratification parameters as the unprocessed baseline. This ensured experimental consistency and comparable class distributions across all model configurations. The resulting dataset, incorporating both integrity validation and targeted linguistic preprocessing, provided a cleaner yet emotionally representative foundation for feature extraction and model training. It therefore established a robust experimental basis for analysing how preprocessing choices influence classification performance across different algorithms.

VI. RESULTS

All experiments were implemented and documented in separate Jupyter (.ipynb) notebooks. Three classification algorithms which are Naïve Bayes, Linear SVM, and Neural Network (MLP) which were applied to each dataset. Specifically, two datasets

were used: the *Covid News Articles* dataset (evaluated on both raw and preprocessed text) and the *Depression Reddit* dataset (evaluated similarly on raw and preprocessed text). The corresponding code notebooks and links are provided in Appendix A.

A. Key observations (Covid New Articles, Raw Dataset).

From the experimental results on the raw dataset for the Covid News Articles Dataset, several important patterns emerged. First, when the data was loaded and the label distributions checked, a clear imbalance among sentiment classes was evident which is structural characteristic sets up particular challenges for the classification models. During hyper-parameter tuning, each algorithm revealed its optimal configuration (for example, max features, n-gram range, alpha with smoothing/regularisation), which permitted the best performance on the validation folds. After tuning, the algorithms were trained on the full training set and evaluated on the hold-out test set, with results reported via confusion matrices and accuracy (among other metrics).

For the neural network classification (MLP), an additional fine-tuning step focused on epochs and learning dynamics. As expected, training accuracy increased with more epochs, but beyond a certain point the validation or test accuracy ceased to improve (or even degraded), and loss started to rise or stagnate signaling overfitting. From this behavior the “best epoch” was selected for final evaluation, balancing high accuracy and minimised test loss.

Finally, the results for each algorithm were compiled and tabulated to compare theoretical expectations (e.g., NB simplicity, SVM linear separation, NN non-linearity) with practical performance outcomes. These observations provide insight into model behavior on high dimensional, imbalanced, multi-class sentiment data and guide interpretation of how preprocessing and feature engineering affect effectiveness.

a) *Raw Dataset (5 classes) results:* These figures indicate that while each of the five sentiment classes is represented, the distribution is moderately imbalanced: the “Positive” and “Negative” classes together account for the majority of samples, while the “Extremely” classes and “Neutral” are comparatively smaller. Class imbalance has important implications for modelling: although overall accuracy may appear strong, it can mask poor performance on under-represented classes because the model can attain high accuracy by focussing on the dominant classes. Evaluation metrics such as macro-F1 or balanced accuracy become more informative in such contexts because they treat each class equally rather than weighting performance toward the majority class. In summary, the dataset is suitable for multi-class sentiment classification with good size and diversity, but care must be taken in model evaluation and tuning to ensure that performance on minority classes is not overshadowed by majority-class accuracy.

TABLE I
LABEL DISTRIBUTION FOR THE FIVE-CLASS SENTIMENT DATASET

Sentiment Class	Count
Positive	12,369
Negative	10,958
Neutral	8,332
Extremely Positive	7,223
Extremely Negative	6,073

After initiating the Naïve Bayes and SVM, the fine tuning is running to find best parameter for high accuracy and show the result. When the adjusting parameters are max features, ngram range and C in SVM or alpha in Naïve Bayes. then the best accuracy in each algorithms are showing as table as confusion matrix.

b) *Imbalanced Dataset:* As you see, the each label is not fairly distributed and it can occur not robust accuracy and result for classification

TABLE II
SUMMARY OF MODEL PERFORMANCE ON RAW DATASET (COVID NEWS ARTICLES, 5 CLASSES)

Model	Accuracy	Precision	Recall	F1
Neural Network (TF-IDF → MLP)	0.6315	0.6424	0.6315	0.6336
Linear SVM (tuned)	0.5785	0.5739	0.5785	0.5751
Naïve Bayes (tuned)	0.4664	0.5084	0.4664	0.4609

c) *Naïve Bayes (TF-IDF unigram/bigram, tuned):* The Naïve Bayes classifier achieved an overall accuracy of 46.6% and macro-F1 of 0.45. Although precision for dominant classes such as *Positive* and *Negative* remained moderate, recall for minority classes (e.g., *Extremely Negative*) dropped below 0.25, indicating substantial bias toward the majority categories. This behaviour reflects the probabilistic model’s assumption of word independence and its limited ability to capture contextual polarity.

TABLE III
COMPARISON OF PER-CLASS PERFORMANCE METRICS ACROSS MODELS (TF-IDF, RAW DATASET)

Model	Sentiment Class	Precision	Recall	F1-score
Naïve Bayes	Extremely Negative	0.66	0.24	0.35
	Extremely Positive	0.62	0.35	0.45
	Negative	0.41	0.53	0.46
	Neutral	0.58	0.45	0.51
	Positive	0.41	0.59	0.49
	Average / Overall	0.54	0.43	0.45
Linear SVM	Extremely Negative	0.61	0.64	0.63
	Extremely Positive	0.65	0.68	0.67
	Negative	0.50	0.46	0.48
	Neutral	0.63	0.72	0.67
	Positive	0.54	0.50	0.52
	Average / Overall	0.59	0.60	0.59
Neural Network (MLP)	Extremely Negative	0.76	0.58	0.66
	Extremely Positive	0.73	0.68	0.70
	Negative	0.54	0.65	0.59
	Neutral	0.69	0.62	0.65
	Positive	0.59	0.62	0.60
	Average / Overall	0.66	0.63	0.64

d) *Linear SVM (TF-IDF unigram, tuned)*.: The Linear SVM outperformed Naïve Bayes with accuracy 57.8% and macro-F1 0.59. It displayed consistent balance between precision and recall, particularly for *Extremely Positive* and *Neutral* classes, while maintaining reasonable separation between neighboring sentiment levels. Its linear decision boundary efficiently handled the sparse TF-IDF representation, though some overlap remained between *Negative* and *Extremely Negative* samples.

e) *Neural Network (TF-IDF $\rightarrow$ MLP)*.: The MLP achieved the best performance overall with accuracy 63.2% and macro-F1 0.64. Its hidden layers captured non-linear feature interactions between words, enabling improved detection of subtle sentiment polarity, particularly for the extreme categories. Early stopping at epoch 7 prevented overfitting while maintaining high validation accuracy (≈ 0.82). The results demonstrate the MLP’s superior capability to generalise on high-dimensional, imbalanced textual data.

f) *Summary Result (Unprocessed)*: Comparing across algorithms, a consistent ranking was observed: **Neural Network** > **SVM** > **Naïve Bayes**. Naïve Bayes, while fast and interpretable, underperformed on minority classes due to its independence assumption. SVM provided a balanced trade-off between bias and variance, while the Neural Network delivered the best overall generalisation through non-linear feature learning. All models showed difficulty distinguishing adjacent sentiment levels (e.g., Neutral vs. Positive), confirming linguistic overlap within the dataset. Performance variations also reflected the influence of class imbalance, where frequent classes inflated global accuracy but depressed macro-F1.

Overall, the Neural Network achieved the most stable performance across folds, while SVM remained a strong linear baseline. These findings are consistent with theoretical expectations from text classification: linear models excel with sparse features, whereas non-linear architectures further enhance feature interaction and class separation.

B. Key observations (Covid New Articles, Pre-processed Dataset).

After text normalisation (contractions, emoji placeholders, hashtag splitting, negation scope, stopwords with negators kept, lemmatisation) and label mapping from five to three classes (Positive/Neutral/Negative), models were tuned and evaluated on a fixed hold-out test set. The 3 class setting reduces label sparsity and consolidates “extreme” sentiment into their parent polarities, which typically improves separability and stability. Class imbalance remains (Positive and Negative dominate Neutral), so macro-averaged metrics remain important.

a) *Class distribution and model performance (3 classes)*.: After mapping from 5 to 3 labels, the dataset remains imbalanced, and the following summarises both the overall class distribution and model performance:

TABLE IV
OVERALL CLASS DISTRIBUTION AND MODEL PERFORMANCE SUMMARY (COVID NEWS ARTICLES, 3 CLASSES)

(a) Class Distribution			(b) Model Performance				
Class	Count	%	Model	Acc.	Prec.	Rec.	F1
Positive	19,592	43.3	Linear SVM	0.8228	0.8217	0.8228	0.8217
Negative	17,031	37.7	Neural Net	0.7899	0.7913	0.7899	0.7904
Neutral	8,332	18.4	Naïve Bayes	0.6901	0.6920	0.6901	0.6726
Total	44,955	100.0					

TABLE V
COMPARISON OF PER-CLASS PERFORMANCE METRICS ACROSS MODELS (PREPROCESSED, 3-CLASS DATASET)

Model	Sentiment Class	Precision	Recall	F1-score
Naïve Bayes	Negative	0.73	0.75	0.74
	Neutral	0.69	0.26	0.38
	Positive	0.66	0.79	0.72
	Average / Overall	0.69	0.60	0.61
Linear SVM	Negative	0.84	0.85	0.84
	Neutral	0.76	0.68	0.72
	Positive	0.81	0.86	0.84
	Average / Overall	0.82	0.79	0.80
Neural Network (MLP)	Negative	0.81	0.82	0.82
	Neutral	0.64	0.67	0.66
	Positive	0.83	0.80	0.82
	Average / Overall	0.76	0.77	0.76

b) Naïve Bayes (TF-IDF, tuned).: The Naïve Bayes classifier achieved a moderate level of performance on the pre-processed three-class dataset, with an overall accuracy of approximately 69% and a macro-F1 score of 0.61. As shown in Table IX, the model demonstrated reasonable performance on the *Negative* and *Positive* categories but struggled considerably with the *Neutral* class, where recall dropped to 0.26. This outcome indicates that while Naïve Bayes effectively captured broad sentiment polarity, it was less capable of distinguishing contextually ambiguous or mixed-sentiment statements. The probabilistic independence assumption likely limited its ability to learn nuanced co-occurrences and phrase-level context, leading to misclassifications particularly near class boundaries.

c) Linear SVM (TF-IDF, tuned).: The Linear SVM obtained the strongest overall results among all algorithms, reaching an accuracy of 82% and a macro-F1 of 0.80. As shown in Table IX, its predictions were consistently balanced across all classes, achieving high precision and recall on both *Positive* and *Negative* sentiments while maintaining solid performance for *Neutral* examples. The SVM’s margin-based optimisation enabled it to construct clear decision boundaries even within the high-dimensional TF-IDF space, leading to robust generalisation and minimal bias toward any particular sentiment polarity. This balanced behavior reflects the strength of linear classifiers in handling sparse textual representations after normalisation.

d) Neural Network (TF-IDF $\rightarrow$ MLP, tuned).: The Multi-Layer Perceptron (MLP) achieved an accuracy of approximately 79% and a macro-F1 of 0.76, positioning it closely behind the SVM. The per-class metrics in Table IX reveal balanced performance across sentiment categories, with the *Neutral* class showing improved recall compared to Naïve Bayes but still lagging behind SVM. The MLP’s non-linear architecture captured interactions between TF-IDF features that linear models could not, enhancing discrimination between strongly polarised statements. However, the limited training data and residual class imbalance slightly constrained its ability to reach the SVM’s precision on borderline cases. Overall, the neural model generalised well and demonstrated that even basic feedforward networks can effectively model sentiment when combined with appropriate regularisation and early stopping.

e) Summary Result (Preprocessed).: Across all models, a clear performance hierarchy was observed: **SVM > Neural Network > Naïve Bayes**. The SVM demonstrated superior consistency and generalisation under the preprocessed 3-class setup, benefiting from cleaner tokenisation, reduced noise, and linear separability in TF-IDF space. The MLP, though slightly lower in accuracy, displayed strong adaptability to complex sentiment cues, particularly when negation and emoji handling preserved contextual polarity. Naïve Bayes improved substantially relative to the raw 5-class setting but remained limited in contextual sensitivity, especially for ambiguous or weakly polarised text. Overall, preprocessing and label consolidation significantly enhanced classification stability and accuracy, validating the importance of linguistic normalisation and sentiment-specific cleaning for reliable text mining results.

C. Key Observations (Depression: Reddit Dataset, Unprocessed)

a) Depression: Reddit, Naïve Bayes.: Based on the output classification table (TABLE VII), the Naïve Bayes model achieved an accuracy of nearly 90% (0.8966). The classification report shows strong performance for detecting depression (class 1), with high recall (0.95) indicating it identifies most actual depression cases, and good precision (0.86) meaning its positive predictions are largely accurate. It also performs well in identifying non-depression cases (class 0) with very high precision (0.95) and good recall (0.84). Overall, the model provides a solid baseline for this text classification task.

b) Depression: Reddit, Regularised Neural Network (MLP).: Compared to the Naïve Bayes model, the Regularised Neural Network (TABLE VIII) shows a significant improvement in overall accuracy, achieving 94.31% compared to Naïve Bayes’ 89.66%. While Naïve Bayes had a slightly higher recall for detecting depression (0.95 vs 0.89), the Regularised Neural Network displays perfect precision (1.00) for this class, meaning it makes no false positive predictions for depression. Furthermore, the Regularised Neural Network perfectly identifies all non-depressed comments with a recall of 1.00, whereas Naïve Bayes only

TABLE VI
MODEL PERFORMANCE ON UNPROCESSED DEPRESSION REDDIT DATASET (BINARY CLASSIFICATION)

Model	Precision	Recall	F1-score	Accuracy
Naïve Bayes	0.91	0.90	0.90	0.8966
Regularised MLP Neural Network	0.95	0.95	0.94	0.9431
CNN	0.96	0.96	0.96	0.9619

achieved 0.84 recall for this class. The higher F1-scores for both classes in the Regularised Neural Network indicate a better overall balance between precision and recall, making it a more effective model despite the minor trade-off in depression recall.

c) Depression: Reddit, CNN: The CNN model (TABLE IX) achieved the highest overall accuracy at 96.19%, surpassing both Naïve Bayes (89.66%) and the Regularised Neural Network (94.31%). Looking at the classification report, the CNN shows balanced performance with precision, recall, and F1-scores all around 0.96 for both depression and non-depression classes. Compared to Naïve Bayes, the CNN is significantly better across all metrics. When compared to the Regularised Neural Network, the CNN has slightly lower precision and recall for the non-depression class but achieves a better balance between precision and recall for both classes, resulting in a higher overall accuracy and slightly better F1-scores. Based on the training history, the CNN showed signs of slight overfitting as the validation loss started to increase after epoch 5, seen in Appendix B, while the training loss continued to decrease, but early stopping helped to mitigate this.

d) Summary Result (Unprocessed): Summarising the results, the CNN emerged as the best performer with the highest overall accuracy (96.19%), demonstrating strong and balanced precision and recall for both depression and non-depression detection, likely due to its ability to identify key phrases and patterns in the text. The Regularised Neural Network followed closely with 94.31% accuracy, showing excellent precision for depression detection and perfect recall for non-depression, benefiting from regularisation to prevent overfitting. The Naïve Bayes model served as a solid baseline but was the worst performer with 89.66% accuracy; while it had good recall for depression, its assumption of word independence limits its ability to capture complex language patterns compared to the neural network models.

D. Key Observation (Depression: Reddit Dataset, preprocessed)

a) Depression: Reddit Dataset, Naïve Bayes: The fine-tuned Naïve Bayes model (TABLE X) shows improved performance on the test set compared to the previous Naïve Bayes models, achieving a higher accuracy and F1-score for the depression class. This improvement is attributed to the fine-tuning process which involved cross-validated hyperparameter tuning. The use of hyperparameters, seen in Appendix C, like `min_df`, `max_df`, `alpha`, and `fit_prior` within the `TfidfVectorizer` and `MultinomialNB` allowed the model to better capture relevant features and generalise to unseen data. The preprocessing steps, including cleaning data issues and using custom stop words, likely contributed by providing a cleaner and more focused text representation for the model to learn from.

TABLE VII
MODEL PERFORMANCE ON PREPROCESSED DEPRESSION REDDIT DATASET (BINARY CLASSIFICATION)

Model	Precision	Recall	F1-score	Accuracy
Naïve Bayes	0.92	0.91	0.91	0.9066
Regularised MLP Neural Network	0.97	0.97	0.97	0.9717
CNN	0.96	0.96	0.96	0.9599

b) Depression: Reddit Dataset, Regularised Neural Network: The fine-tuned Regularised Neural Network (TABLE XI) shows strong performance on the test set, achieving high accuracy and an excellent F1-score for the depression class. This is attributed to the fine-tuning process, which included using a validation set, regularisation, and learning rate reduction to improve generalisation. The classification report further highlights the model's effectiveness for the depression class, with a precision of 0.99, and a recall of 0.95. Overall, the preprocessed Regularised Neural Network significantly outperforms the fine-tuned Naïve Bayes model, showcasing the neural network's capability to capture complex patterns for accurate depression detection.

c) Depression: Reddit Dataset, CNN: The preprocessed CNN model achieved a test accuracy of about 96.0% and an F1-score of 96.0% for the depression class. While it shows good performance, the training history reveals a gap between training and validation accuracy seen in Appendix D, suggesting some degree of overfitting despite the use of dropout and early stopping. Compared to the fine-tuned Naïve Bayes model, the preprocessed CNN shows a slight improvement in overall accuracy and F1-score for the depression class. However, the fine-tuned Regularised Neural Network appears to be the best performing model among the three, with a higher overall accuracy and F1-score for the depression class, and better handling of overfitting due to its more comprehensive regularisation techniques and a dedicated validation split during training.

d) Summary Result (Preprocessed): Comparing the fine-tuned models, the Regularised Neural Network emerged as the top performer after preprocessing and fine-tuning, achieving approximately 97.2% accuracy and a 97.0% F1-score for the depression class. This represents a significant improvement over the unprocessed models and slightly surpasses the preprocessed CNN (96.0% accuracy, 96.0% F1 for depression), which showed signs of minor overfitting despite preprocessing and basic regularisation. The fine-tuned Naïve Bayes model, while also improved by preprocessing and hyperparameter tuning, remained the lowest performer at around 90.7% accuracy, demonstrating the limitations of its simpler approach compared to the neural networks. The preprocessing and fine-tuning steps consistently improved the performance of all models, with the preprocessed Regularised Neural Network showing the most robust generalisation.

VII. DISCUSSION

A. Covid New Articles Dataset

In transitioning from the raw five-class dataset to the three-class version, we observed improved class balance and model performance. The original five categories were heavily weighted toward “Positive” and “Negative” sentiments, which increased the risk of the model neglecting the fewer “Extremely” sentiment cases. By consolidating classes into Positive, Negative, and Neutral and applying text preprocessing (e.g., negation scope tagging, hashtag splitting, emoji normalisation), we reduced sparsity in the minority classes and enriched feature representation. As a result, the dataset became more evenly distributed ($\approx 44\%$ Positive, 38% Negative, 18% Neutral) and enabled fairer learning across classes. This effect, combined with the more discriminative features, contributed to higher accuracy and more balanced precision/recall across sentiment classes.

TABLE VIII
SUMMARY OF SENTIMENT DATASET DISTRIBUTIONS AND MODEL PERFORMANCE (SPACE-OPTIMISED LAYOUT)

(A) Label Distributions				(B) Model Performance (5-Class Raw)						
Dataset	Sentiment Class	Count	%	Model	Sentiment Class	Precision	Recall	F1-score	Accuracy	
Raw (5-Class)	Positive	12,369	27.5	Naïve Bayes	Extremely Negative	0.66	0.24	0.35	0.4664	
	Negative	10,958	24.4		Extremely Positive	0.62	0.35	0.45		
	Neutral	8,332	18.5		Negative	0.41	0.53	0.46		
	Extremely Positive	7,223	16.1		Neutral	0.58	0.45	0.51		
	Extremely Negative	6,073	13.5		Positive	0.41	0.59	0.49		
	Total	44,955	100.0		Average / Overall	0.54	0.43	0.45		
Preprocessed (3-Class)	Positive	19,592	43.6	Neural Network	Extremely Negative	0.76	0.58	0.66	0.6315	
	Negative	17,031	37.9		Extremely Positive	0.73	0.68	0.70		
	Neutral	8,332	18.5		Negative	0.54	0.65	0.59		
	Total	44,955	100.0		Neutral	0.69	0.62	0.65		
			Positive		0.59	0.62	0.60			
(C) Model Performance (3-Class Preprocessed)					Average / Overall	0.66	0.63	0.64		
Model	Sentiment Class	Precision	Recall	F1-score	Accuracy	Linear SVM	Extremely Negative	0.61	0.64	0.63
Naïve Bayes	Negative	0.73	0.75	0.74	0.6901		Extremely Positive	0.65	0.68	0.67
	Neutral	0.69	0.26	0.38			Negative	0.50	0.46	0.48
	Positive	0.66	0.79	0.72			Neutral	0.63	0.72	0.67
	Average / Overall	0.69	0.60	0.61			Positive	0.54	0.50	0.52
Neural Network	Negative	0.81	0.82	0.82	0.7899	Average / Overall	0.59	0.60	0.59	
	Neutral	0.64	0.67	0.66						
	Positive	0.83	0.80	0.82						
	Average / Overall	0.76	0.77	0.76						
Linear SVM	Negative	0.84	0.85	0.84	0.8228					
	Neutral	0.76	0.68	0.72						
	Positive	0.81	0.86	0.84						
	Average / Overall	0.81	0.79	0.80						

The results demonstrate clear performance improvements following preprocessing and label consolidation. On the raw 5-class dataset the highest accuracy achieved was 63.15% with the MLP, while the lowest was 46.64% with Naïve Bayes (Table VIII). Performance differed considerably between classes: for example, the “Extremely Negative” category received a recall as low as 24% under Naïve Bayes (Table VIII), indicating significant difficulty in capturing minority extreme classes. After consolidating to a 3-class scheme and applying richer text preprocessing, performance rose substantially: the tuned Linear SVM reached 82.28% accuracy and a macro-F1 of 0.80 (Table VIII). In the per-class breakdown (Table VIII), the Neutral class improved markedly (recall ≈ 0.68), reducing the class-imbalance bias seen earlier. These changes reflect three consistent patterns: (a) simpler probabilistic models (Naïve Bayes) struggle most with sparsely populated or ambiguous classes, (b) linear models (SVM) benefit strongly from cleaned, normalised features and label consolidation, and (c) neural networks bridge the gap by capturing interaction effects, yet still lag behind the best linear model when data is moderately sized and sparse. An aberration worth noting is that even after label consolidation the Neutral class remains the weakest link for all models, suggesting that ambiguous sentiment remains inherently difficult to classify. Overall, the best-performing algorithm was the Linear SVM applied to the preprocessed dataset, combining the most favourable balance of accuracy, recall/precision across classes, and

computational simplicity. These findings emphasise the value of both preprocessing and label engineering as foundational steps in text classification tasks.

B. Depression: Reddit Dataset

Critically evaluating the three algorithms, the fine-tuned preprocessed Regularised Neural Network showed the best overall performance, achieving high precision (99%) and recall (95%) for the depression class and the highest overall accuracy (97.2%). This highlights the power of neural networks with appropriate regularisation and a validation split in capturing complex patterns in text data. The preprocessed CNN also performed well (96.0% accuracy), particularly in identifying depression cases (96% precision and recall), likely due to its ability to detect local patterns and phrases, although it showed some overfitting. In contrast, the fine-tuned Naïve Bayes model, despite preprocessing improvements, remained the weakest performer (90.7% accuracy), underscoring its limitation in handling word dependencies. A consistent pattern observed is the general improvement in performance across all models after preprocessing and fine-tuning, with the neural networks benefiting more significantly. While all models showed good recall for the depression class, the fine-tuned preprocessed Regularised Neural Network’s balanced high precision and recall, coupled with its robust handling of overfitting through dedicated validation and regularisation, made it the most effective algorithm for this dataset. The difference in performance between the unprocessed and processed models, especially for the neural networks, emphasises the importance of data preparation and model tuning for this task.

TABLE IX
COMPARISON OF PER-CLASS PERFORMANCE METRICS ACROSS MODELS (PREPROCESSED, BINARY CLASSIFICATION)

Model	Class	Precision	Recall	F1-score
Naïve Bayes	Class 0	0.97	0.84	0.90
	Class 1	0.86	0.97	0.91
	Accuracy / Overall		0.9066	
Regularised Neural Network (MLP)	Class 0	0.96	0.99	0.97
	Class 1	0.99	0.95	0.97
	Accuracy / Overall		0.9717	
CNN	Class 0	0.96	0.97	0.96
	Class 1	0.96	0.95	0.96
	Accuracy / Overall		0.9599	

C. Evaluation of both results on both datasets

In the multi-class COVID News Articles, the Linear SVM achieved the highest accuracy of about 82% on the preprocessed data, which outperforms both Naïve Bayes and Neural Network models. This reflects SVM’s strength in handling high-dimensional TF-IDF features and overfitting. The Neural Network (MLP) performed well with an accuracy of around 79% but required more tuning than the others. Naïve Bayes fell behind in terms of performance due to its limited contextual awareness and sensitivity to class imbalances.

In contrast, the Depression: Reddit Dataset produced significantly higher overall accuracies across all models. The Regularised Neural Network achieved the best performance of around 97%, followed by CNN with a performance of around 96%. Both of these models were effective in capturing emotional expressions in semantic patterns. Naïve Bayes, though the most efficient, remained the weakest performer, relative to the other algorithms in the Depression: Reddit dataset, highlighting the limitations of probabilistic assumptions for nuanced emotional text.

Preprocessing consistently improved results across both datasets. In the COVID dataset, label consolidation and text normalisation enhanced class balance and interpretability, while in the Reddit dataset, targeted cleaning and selective stopword retention preserved affective cues critical for depression detection. Neural models benefited more from these refinements, though linear models remained more stable and generalisable.

Overall, the results confirm that performance depends less on algorithmic complexity than on data quality and representation. Linear SVMs are most effective for structured sentiment analysis, while neural models excel in emotionally rich, context-driven text. Preprocessing proved essential in both cases, reinforcing its role as the foundation of reliable text classification outcomes.

VIII. CONCLUSION AND REFLECTION

With 3 labels (Positive/Neutral/Negative) plus simple text cleaning, a tuned Linear SVM was the most reliable choice. It beat Naïve Bayes and a small MLP on accuracy and balance. In this study, we evaluated three classification algorithms which are Naïve Bayes, Linear SVM, and a simple feed-forward neural network (MLP) on a multi-class sentiment dataset of Tweets and Reddits. The results indicate clear performance improvements when moving from the original five-class task to a three-class (Positive, Negative, Neutral) label space, alongside enhanced text preprocessing. Specifically, in the original raw data the best

accuracy (63%) was achieved by the neural network, while Naïve Bayes managed only 47%. After consolidating extreme sentiment labels and applying normalisation and feature engineering (such as negation tagging, contraction expansion, emoji, and hashtag processing), the tuned SVM achieved 82% accuracy and the neural network 79%.

Naïve Bayes is super fast but struggles on Neutral. SVM trains in about a minute or two on CPU and is cheap to run best overall. MLP/CNN need more tuning and a bit more time, but didn't beat SVM here.

Our results could be biased for a few simple reasons. What works here might not transfer perfectly to other types of text. Some posts are genuinely unclear, which makes labeling hard and especially hurts the neutral class. Our cleaning rules (like how we handle emojis, hashtags, or negations) may not fit other platforms or datasets the same way. Running multiple different splits would give tighter, more reliable estimates in comparison to our one-test split.

The algorithm-dataset interplay also reveals some guiding patterns. Linear SVM proved especially suitable for high-dimensional, sparse text representations (TF-IDF) with moderate volume which is confirming prior research that SVMs excel in text classification when features are well engineered.

Naïve Bayes, despite its speed and simplicity, consistently under-performed on minority and ambiguous classes (particularly "Neutral"), highlighting its limitations when contextual nuance matters. The neural network, while competitive, required more tuning and computational effort, and did not surpass the SVM in this setting suggesting a trade-off between modeling complexity and practical returns for this dataset size and feature type.

By contrast, the Depression Reddit dataset which simpler in structure but richer in emotional depth favoured neural models. The Regularised Neural Network achieved the highest accuracy around 97%, demonstrating strong generalisation and the ability to detect subtle emotional cues and context-dependent phrases such as "feel hopeless" or "tired of life." The CNN also performed well, around 96%, but showed slight overfitting despite early stopping and dropout regularisation. Naïve Bayes remained computationally efficient (training in seconds compared to minutes for the neural networks) but lacked contextual awareness, resulting in lower accuracy at around 91%. These findings suggest that linear models such as SVM are optimal for structured, high-dimensional sentiment data, while regularised neural architectures perform best for emotionally expressive, sequential text.

Reflection on preprocessing reveals that merging under-represented classes reduced category sparsity and improved sample counts per class. More importantly, text-level transformations enriched semantic signal and enabled more coherent feature vectors. Without resampling or oversampling, the class imbalance was partially mitigated through more meaningful features and more evenly distributed label frequencies ($\approx 44\%$ Positive, $\approx 38\%$ Negative, $\approx 18\%$ Neutral). This underlines the importance of preprocessing and label engineering as foundational steps, especially in social media or noisy-text sentiment tasks.

If we were to repeat this project, we would expand several aspects across both datasets. First, we would explore embedding-based representations (e.g., Word2Vec, GloVe, or transformers) to better capture non-linear and semantic relationships beyond TF-IDF for improved sentiment and emotion classification. Second, we would record computational costs such as training time, inference speed, and memory usage to evaluate efficiency trade-offs, as neural models on the Depression Reddit dataset required significantly longer training than the linear models on the COVID dataset. Third, we would apply targeted techniques to improve weaker classes by using class-weighting or focal loss for the Neutral class in the COVID dataset, and regularisation or data augmentation to strengthen generalisation on subtle depressive cues in the Reddit dataset.

What was done well was maintaining consistent comparisons across both datasets under the same split, preprocessing, and feature regime, ensuring fair and reproducible evaluation. Rigorous cross-validation enhanced the reliability of results. The main limitation was restricting experiments to TF-IDF and shallow architectures; future work should explore richer embedding-based and transformer models to capture deeper contextual and emotional features across datasets.

What was done well was the structured and fair comparison of models under identical experimental conditions, the clear demonstration of how preprocessing influences model effectiveness, and the establishment of reliable evaluation baselines across datasets. The project successfully illustrated that strong text classification results depend less on algorithmic sophistication and more on appropriate model-data alignment and rigorous preprocessing. Overall, the Linear SVM proved most effective for structured, topic-driven sentiment analysis, while regularised neural networks excelled in detecting emotional and psychological language patterns. These findings highlight how methodological design, data quality, and thoughtful preprocessing collectively determine the success and generalisability of machine learning models in text classification.

We learned a few clear things from this project. Most of the improvement came from smarter labels and basic text cleaning, not from using bigger or fancier models. A well-tuned Linear SVM is still a strong, reliable baseline for short, sparse text like tweets and Reddit posts. And when making choices, it's better to focus on the class that matters most (here, Neutral) and on real-world limits like speed, cost, and explainability—not just the headline accuracy number.

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APPENDIX

A. Python Code File Links

- https://colab.research.google.com/drive/15WQqQKTTI_hFLTa2y2w_HscIPQh0h6TT?usp=sharing
COMP723_Covid_New_Articles_Dataset.ipynb: NB/SVM/MLP on raw TF-IDF.
- <https://colab.research.google.com/drive/1en0tIOcwqPz-Hlt4mNhvCEjGQCW52UKw?usp=sharing>
COMP723_Covid_New_Articles_Dataset_preprocessed.ipynb: NB/SVM (GridSearchCV) + MLP (random search) on preprocessed text.
- <https://colab.research.google.com/drive/1jy6BaAMqcPQIPBHI8ycXpK8z4rIex-h#scrollTo=PgBGENgweYbn>
COMP723_Depression_Dataset.ipynb
- <https://colab.research.google.com/drive/1ogx3lzo3qV7EFGswEhkVOQXAANhCGhF2>
COMP723_Preprocessed_Depression_Dataset.ipynb

B. Unprocessed CNN Epoch Result

```
Epoch 1/15
194/194 ————— 7s 26ms/step - accuracy: 0.9104 - loss: 0.2550 - val_accuracy: 0.9438 - val_loss: 0.1617
Epoch 2/15
194/194 ————— 5s 24ms/step - accuracy: 0.9699 - loss: 0.0991 - val_accuracy: 0.9573 - val_loss: 0.1269
Epoch 3/15
194/194 ————— 4s 20ms/step - accuracy: 0.9783 - loss: 0.0776 - val_accuracy: 0.9476 - val_loss: 0.1517
Epoch 4/15
194/194 ————— 4s 22ms/step - accuracy: 0.9850 - loss: 0.0579 - val_accuracy: 0.9295 - val_loss: 0.2162
Epoch 5/15
194/194 ————— 5s 27ms/step - accuracy: 0.9901 - loss: 0.0417 - val_accuracy: 0.9619 - val_loss: 0.1219
Epoch 6/15
194/194 ————— 4s 20ms/step - accuracy: 0.9903 - loss: 0.0379 - val_accuracy: 0.9483 - val_loss: 0.1725
Epoch 7/15
194/194 ————— 4s 20ms/step - accuracy: 0.9943 - loss: 0.0294 - val_accuracy: 0.9606 - val_loss: 0.1346
Epoch 8/15
194/194 ————— 6s 29ms/step - accuracy: 0.9940 - loss: 0.0244 - val_accuracy: 0.9509 - val_loss: 0.1929
Epoch 8: early stopping
Restoring model weights from the end of the best epoch: 5.
49/49 ————— 0s 7ms/step
```

C. Naïve Bayes Preprocessed Code

```
## NB 2.0 fine tuned
def naive_bayes_processed(X_train, X_test, y_train, y_test):

    ...

    # Vectorizer + NB in a pipeline
    pipe = Pipeline([
        ("tfidf", TfidfVectorizer(
            strip_accents="unicode",
            sublinear_tf=True,
            norm="l2",
            ngram_range=(1,1),
            max_features=20000
        )),
        ("clf", MultinomialNB())
    ], memory=memory)

    # Smaller, sensible grid
    param_grid = {
        "tfidf__min_df": [2, 5],
        "tfidf__max_df": [0.90, 0.95],
        "clf__alpha": [1e-3, 1e-2, 1e-1, 1.0],
        "clf__fit_prior": [True, False],
    }

    ...
```

```
# Run NB 2.0 (FAST) with your current splits
nb_model, vectorizer_processed, nb_pred_processed = naive_bayes_processed(
    X_train, X_test, y_train, y_test
)
```

D. Preprocessed CNN Accuracy/Loss Graph

